Volume

Exam code: 0607

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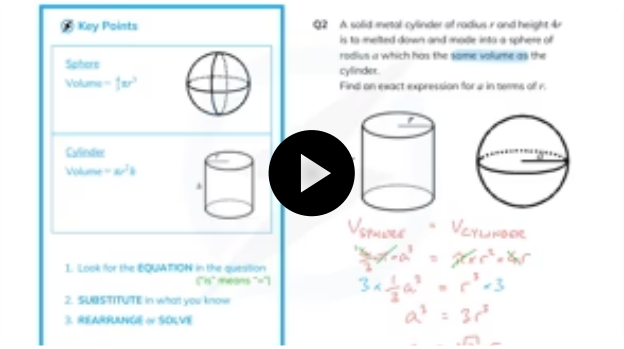

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Help me understand

Volume

What is volume?

- The **volume** of a 3D shape is a measure of how much **space** it takes up
- You need to be able to calculate the volumes of a number of **common 3D shapes**, including:
  - Cubes and cuboids
  - Prisms
  - Pyramids
  - Cylinders
  - Spheres

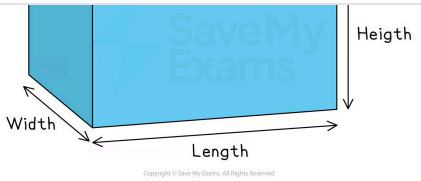
How do I find the volume of a cube or a cuboid?

- A **cube** is a special **cuboid**, where the length, width and height are all of **equal length**
- A **cuboid** is another name for a rectangular-based **prism**
- To find the volume, V, of a **cube** or a **cuboid**, with length, l, width, w, and height, h, use the formula
  - $V = lwh$
  - This formula is **not** given to you in the exam

Stuck on something?

Ask questions and get help with exactly what you're revising.

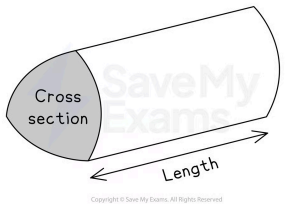
Got it



- You will sometimes see the terms 'depth' or 'breadth' instead of 'height' or 'width'

How do I find the volume of a prism?

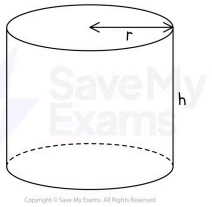
- A **prism** is a 3D object with a **constant cross-sectional area**
- To find the volume, *V*, of a **prism**, with cross-sectional area, *A*, and length, *l*, use the formula
  - $V = Al$
  - This formula is given to you in the exam



- Note that the cross-section can be **any shape**, so as long as you know its **area** and the **length** of the prism, you can calculate its **volume**
  - If you know the **volume** and **length** of the prism, you can calculate the **area** of the **cross-section**

How do I find the volume of a cylinder?

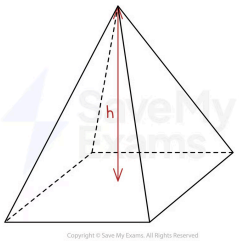
- To calculate the volume, *V*, of a **cylinder** with radius, *r*, and height, *h*, use the formula
  - $V = \pi r^2 h$
  - This formula is given to you in the exam



- Note that a cylinder is similar to a **prism,** its cross-section is a circle with area πr^2, and its length is *h*

How do I find the volume of a pyramid?

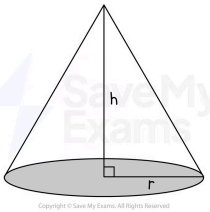
- To calculate the volume, *V*, of a **pyramid** with base area, *A*, and perpendicular height, *h*, use the formula



- The height must be a line from the top of the pyramid that is **perpendicular** to the base
- The **base** of a pyramid could be a square, a rectangle or a triangle

How do I find the volume of a cone?

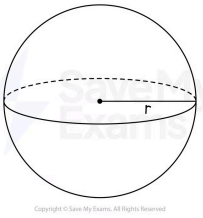
- To calculate the volume, *V*, of a **cone** with base radius, *r*, and perpendicular height, *h*, use the formula
  - $V = \frac{1}{3}\pi r^2 h$
  - This formula is given to you in the exam



- Note that volume formula for a **cone** is similar to a **pyramid**
- The height must be a line from the top of the cone that is **perpendicular** to the base

How do I find the volume of a sphere?

- To calculate the volume, *V*, of a **sphere** with radius, *r*, use the formula
  - $V = \frac{4}{3}\pi r^3$
  - This formula is given to you in your exam



Examiner Tips and Tricks

Worked Example

A cylinder is shown.

The radius, r, is 8 cm and the height, h, is 20 cm.

Calculate the volume of the cylinder, giving your answer correct to 3 significant figures.

Answer:

A cylinder is similar to a prism but with a circular base
The volume of any prism, V, is its base area × height, h, where the base area here is for a circle, πr^2

$$V = \pi r^2 h$$

Substitute $r = 8$ and $h = 20$ into the formula

$$V = \pi \times 8^2 \times 20$$

Work out this value on a calculator

4021.238...

Round the answer to 3 significant figures

4020 cm^3

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